UNDERSTANDING RISING INEQUALITY

Evidence from the World Inequality Report and India · Piketty's r > g · Implications for policy
July 2026

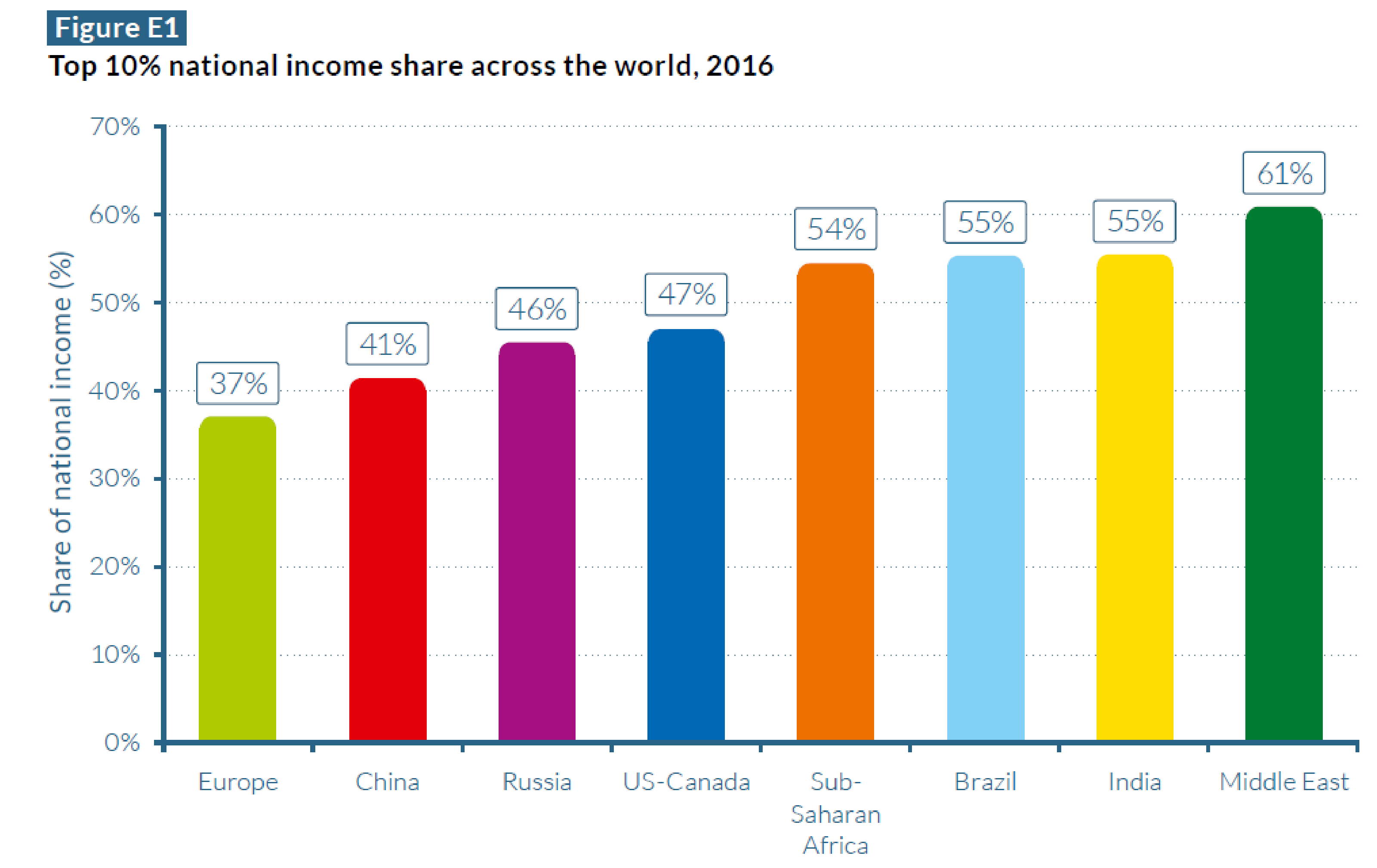
Sources: World Inequality Report 2018 (World Inequality Lab) · Chancel & Piketty, "Indian Income Inequality, 1922–2014" (2017)

PART 1

Is inequality rising — globally and in India?

Building the case with world and India-specific evidence, side by side

Inequality varies hugely across world regions



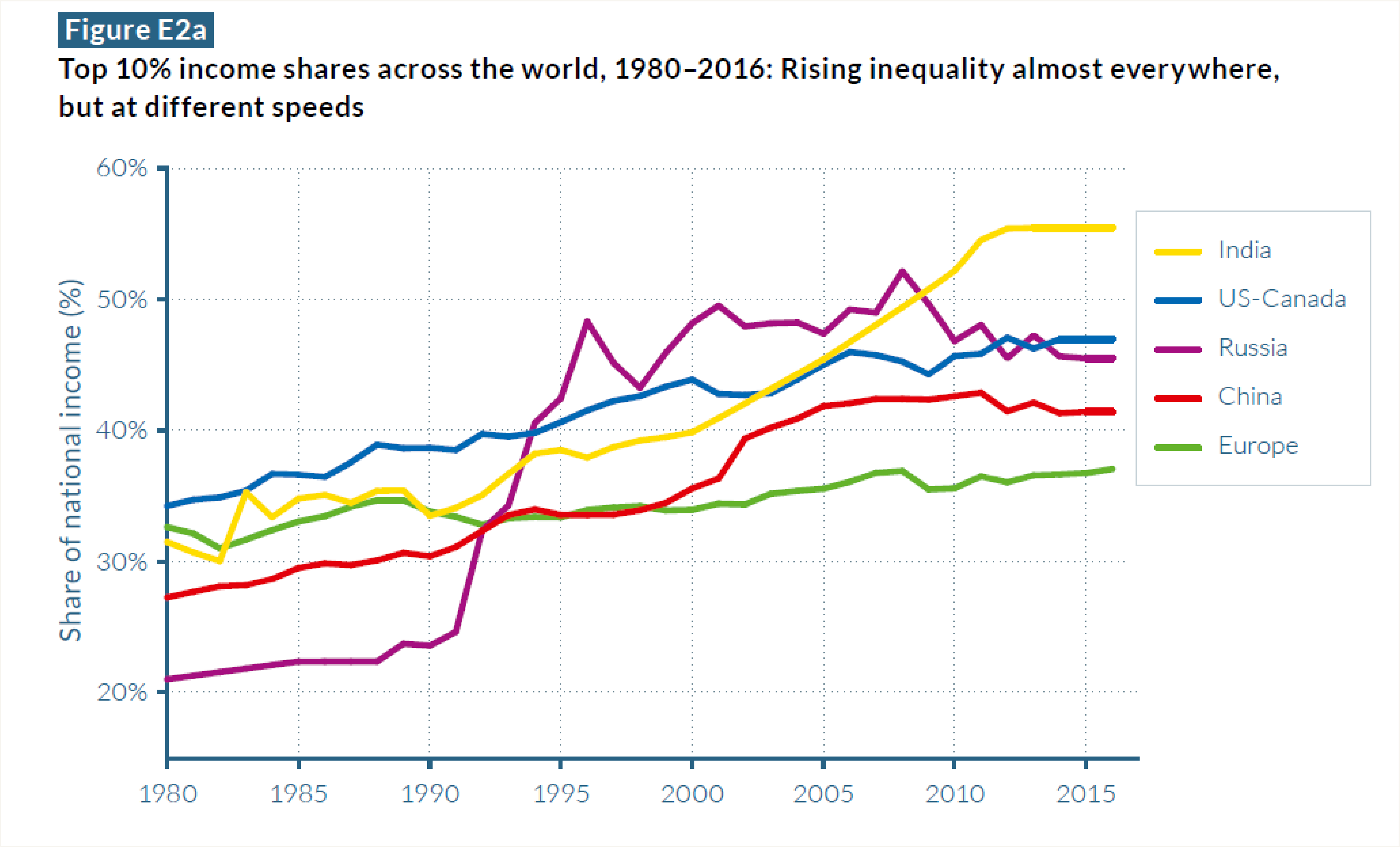
Top 10% national income share, 2016

Europe sits lowest at 37%. The Middle East is highest, at 61%. India and Brazil are close behind sub-Saharan Africa, at 55%.

Countries at similar development levels land in very different places — a first clue that national policy and institutions, not just economic stage, shape inequality.

Source: WID.world (2017), World Inequality Report 2018, Figure E1.

Rising almost everywhere — but at different speeds



Top 10% income share, 1980–2016

Inequality has climbed rapidly in North America, China, India, and Russia since 1980.

Europe's rise has been comparatively moderate — this marks the end of the shared postwar "egalitarian regime" that took different forms across regions.

Source: WID.world (2017), World Inequality Report 2018, Figure E2a.

Wait — GDP or GNP? What are we actually measuring?

GDP

Gross Domestic Product

Value of output produced within a country's borders — regardless of who owns the income. A foreign-owned factory's profit still counts.

GNP / GNI

Gross National Product / Income

Total income earned by a country's citizens/residents, wherever in the world it was earned. Includes income sent home from abroad; excludes profits repatriated by foreign firms.

GNI = GDP + income earned abroad by residents – income earned domestically by foreigners

Why inequality research uses national income, not GDP: Inequality is a question about what people earn, not where production happens. To compare citizen A's income with citizen B's, income must be attributed to persons — which is exactly what national income does and GDP does not. A country with lots of foreign-owned mining could have high GDP, but if the profits leave the country, GNI — and what citizens actually receive — is much lower.

A NATURAL EXPERIMENT

Same starting point, radically different paths: US vs. Western Europe

1980

Top 1% share ~10% in both regions

2016

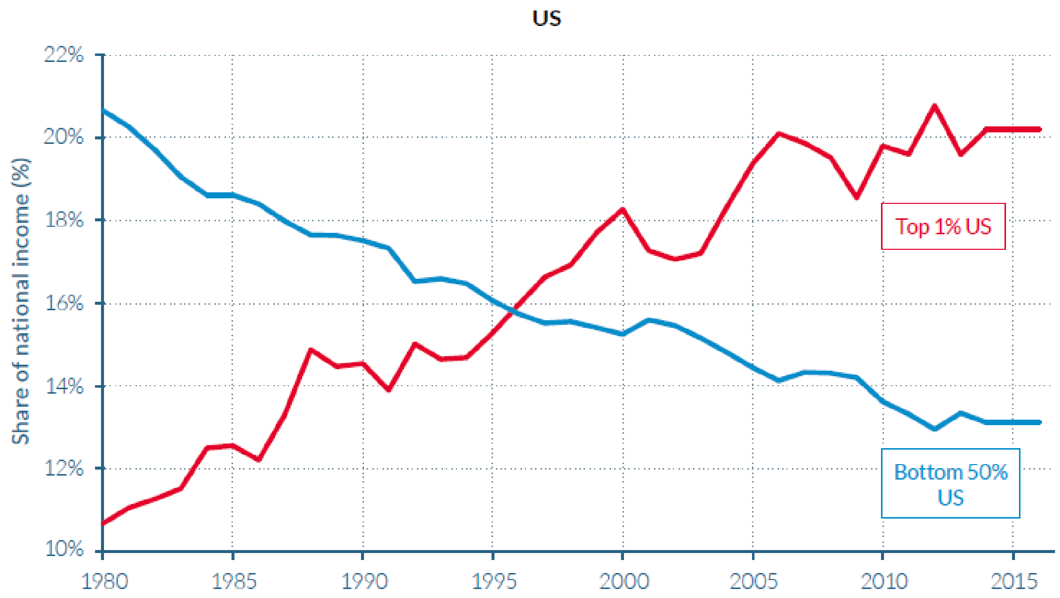
US top 1%: 20%
US bottom 50%: fell to 13%
W. Europe top 1%: 12%
W. Europe bottom 50%: ~22%

Same starting line in 1980. Different tax progressivity, education access, and wage-setting policy explain the split.

Source: WID.world (2017), World Inequality Report 2018, Figure E3.

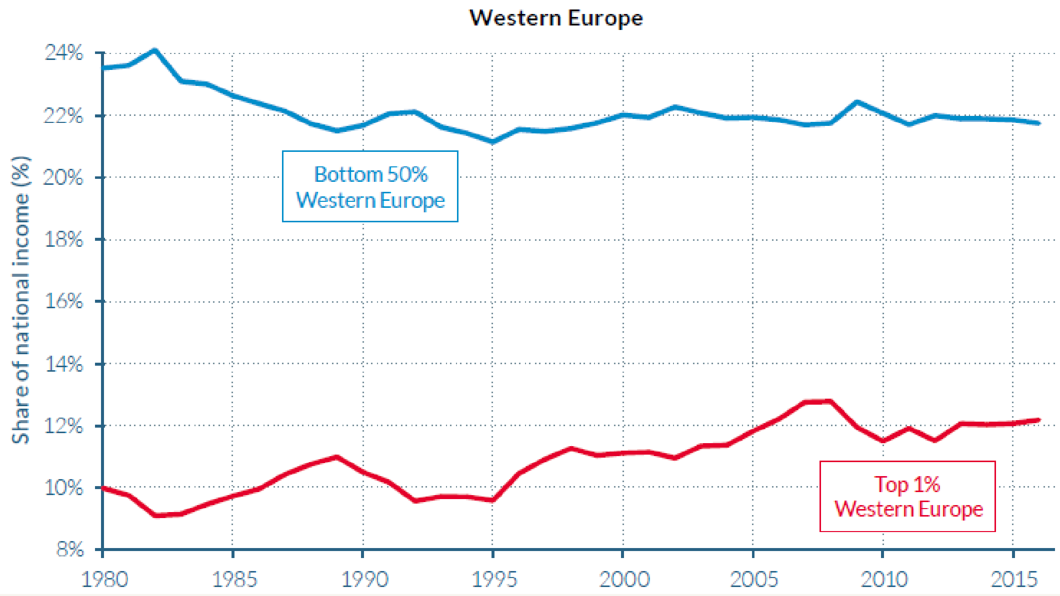
Figure E3

Top 1% vs. Bottom 50% national income shares in the US and Western Europe, 1980–2016: Diverging income inequality trajectories



Source: WID.world (2017). See wir2018.wid.world for data series and notes.

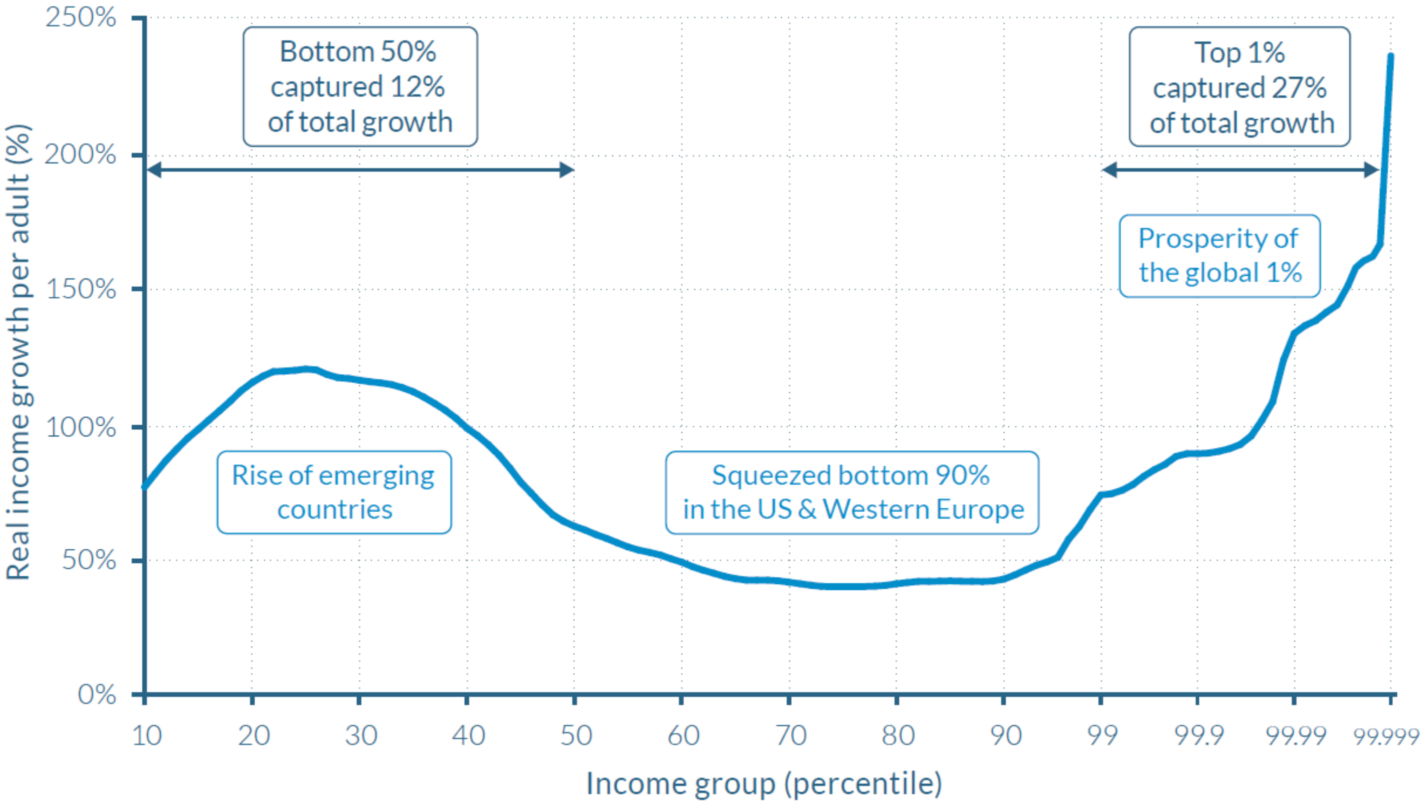
In 2016, 12% of national income was received by the top 1% in Western Europe, compared to 20% in the United States. In 1980, 10% of national income was received by the top 1% in Western Europe, compared to 11% in the United States.



The "elephant curve": who captured global growth, 1980–2016?

Figure E4

The elephant curve of global inequality and growth, 1980–2016



Source: WID.world (2017). See wir2018.wid.world for more details.

On the horizontal axis, the world population is divided into a hundred groups of equal population size and sorted in ascending order from left to right, according to each group's income level. The Top 1% group is divided into ten groups, the richest of these groups is also divided into ten groups, and the very top group is again divided into ten groups of equal population size. The vertical axis shows the total income growth of an average individual in each group between 1980 and 2016. For percentile group p99p99.1 (the poorest 10% among the world's richest 1%), growth was 74% between 1980 and 2016. The Top 1% captured 27% of total growth over this period. Income estimates account for differences in the cost of living between countries. Values are net of inflation.

2x

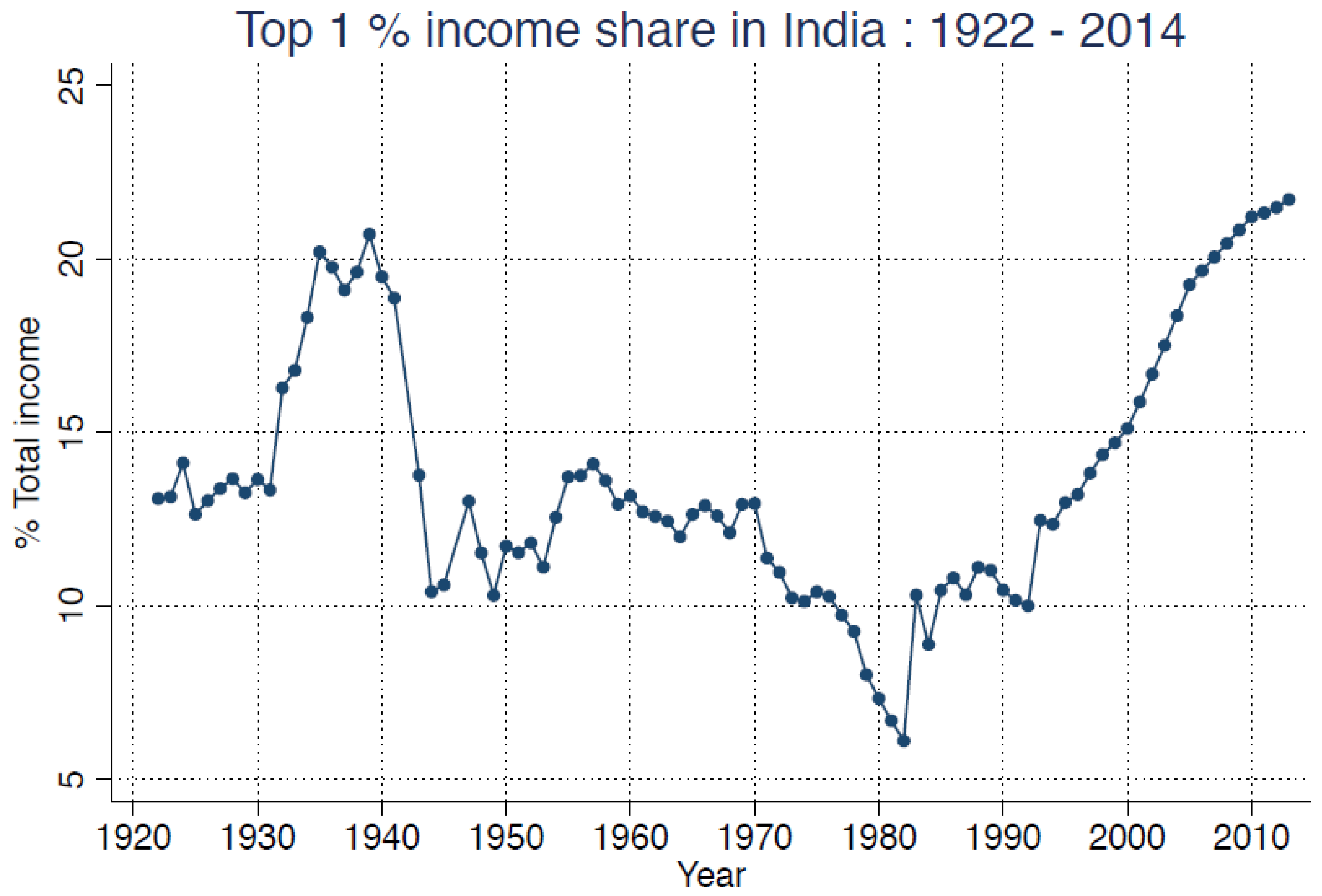
growth captured by global top 1% vs. bottom 50%

The global top 1% captured twice as much growth as the bottom 50%. The "squeezed back" of the elephant is the lower/middle class of rich countries.

Source:

India's top 1% income share, 1922–2014

Figure 6 - Top 1% income share in India, 1922-2014



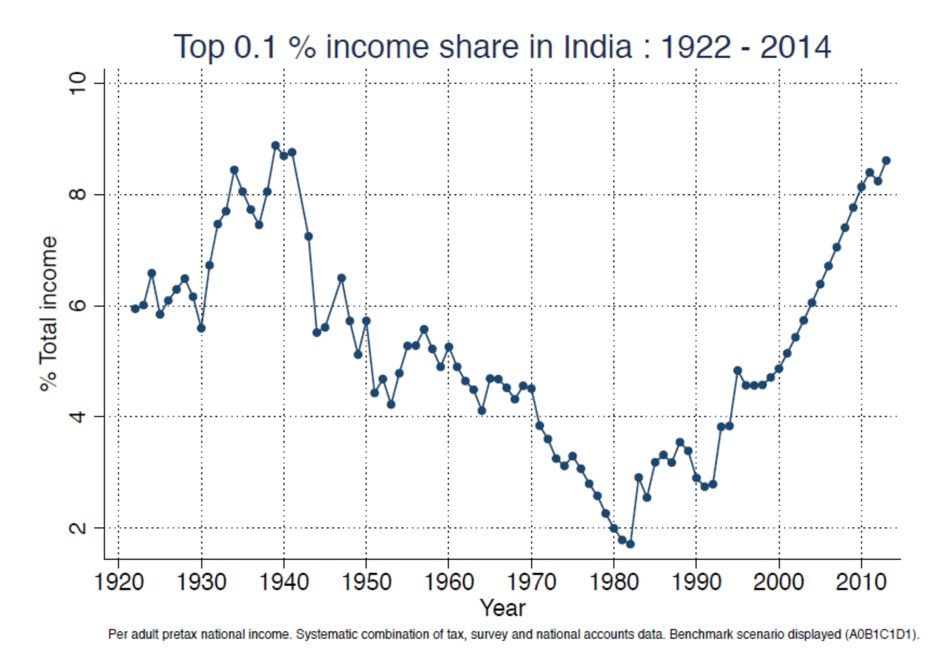
Per adult pretax national income. Systematic combination of tax, survey and national accounts data. Benchmark scenario displayed (A0B1C1D1).

13% → **21%** → **6%** → **22%**
1922 1939 1982 2014

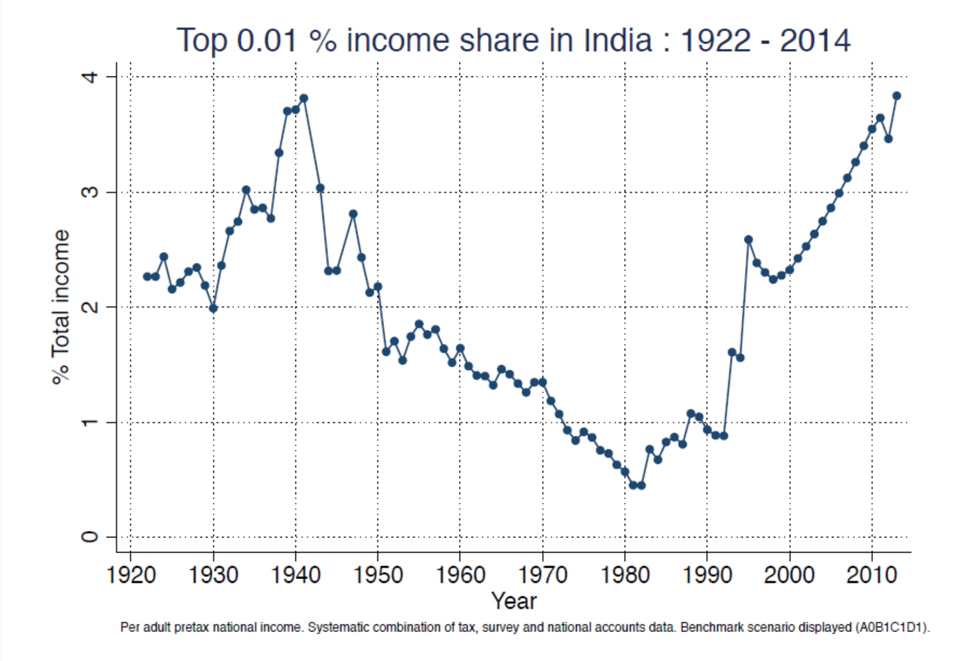
The top 1% share peaked before WWII, collapsed under socialist-era planning and high tax progressivity, then surged past its pre-independence high after 1980s liberalization — now the highest ever recorded.

Source: Chancel & Piketty (2017), "Indian Income Inequality, 1922–2014," Figure 6.

Zooming in: the top 0.1% and top 0.01%



Top 0.1%: 1.7% (1982-83) → 8.6% (2013-14)

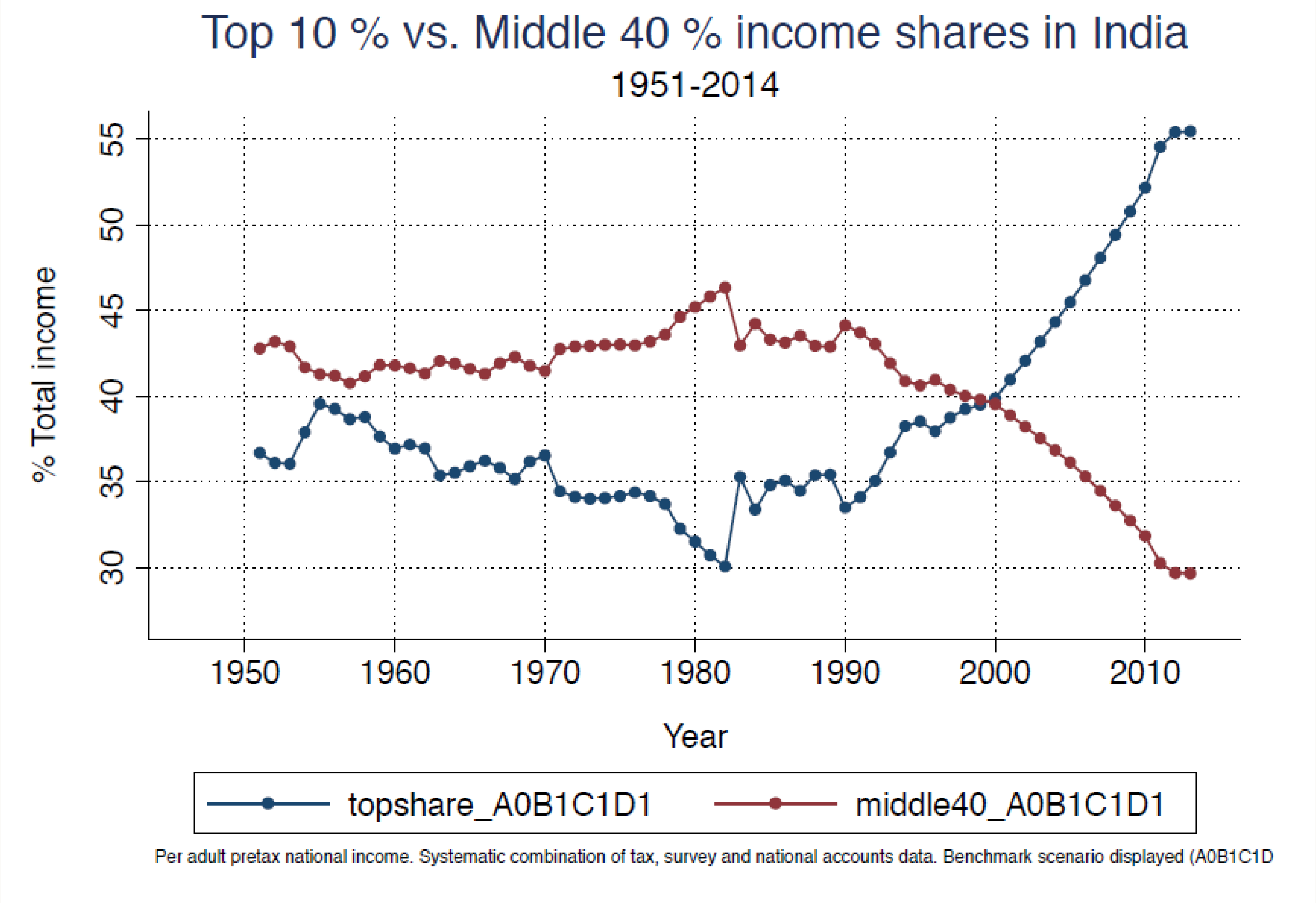


Top 0.01%: 0.4% (1982-83) → 3.8% (2013-14)

Even within the top 1%, gains are concentrated further and further toward the very top — a nearly 10x rise for the top 0.01% since the early 1980s.

Source: Chancel & Piketty (2017), Figures 7 and 8.

The squeezed middle: top 10% vs. middle 40%



46% → 29.6%

Middle 40% share, 1982-83 to 2013-14 — a historic low

This is India's own version of the global elephant curve: as the top 10% share rose, the middle 40% share fell in near mirror-image, just as the global middle class was squeezed between 1980 and 2016.

Source: Chancel & Piketty (2017), Figure 9.

PART 2

Why is this happening? Piketty's r > g

A structural mechanism behind capital-driven inequality

Land, labour, capital — and why capital behaves differently

LAND

Natural resources and space used in production

LABOUR

Human effort — a flow tied to a person. Stop working, income stops.

CAPITAL

Accumulated wealth — factories, financial assets, real estate. Keeps generating income whether or not the owner works.

The key distinction: labour income requires ongoing work. Capital income (rent, dividends, interest, profit) flows independently of effort.

How capital accumulates on its own

1. You own capital (e.g. financial assets, property)
2. Each year, a share of national income flows to you as a return (r)
3. If you save/reinvest part of that return, your capital stock grows
4. A larger capital stock earns a larger absolute return next year

Compounding, not effort

Wealth held by the already-wealthy grows on its own trajectory — separate from how fast wages or the whole economy are growing.

When r > g, capital's share of income rises mechanically

r

Rate of return on capital
(interest, dividends, rent, profit)

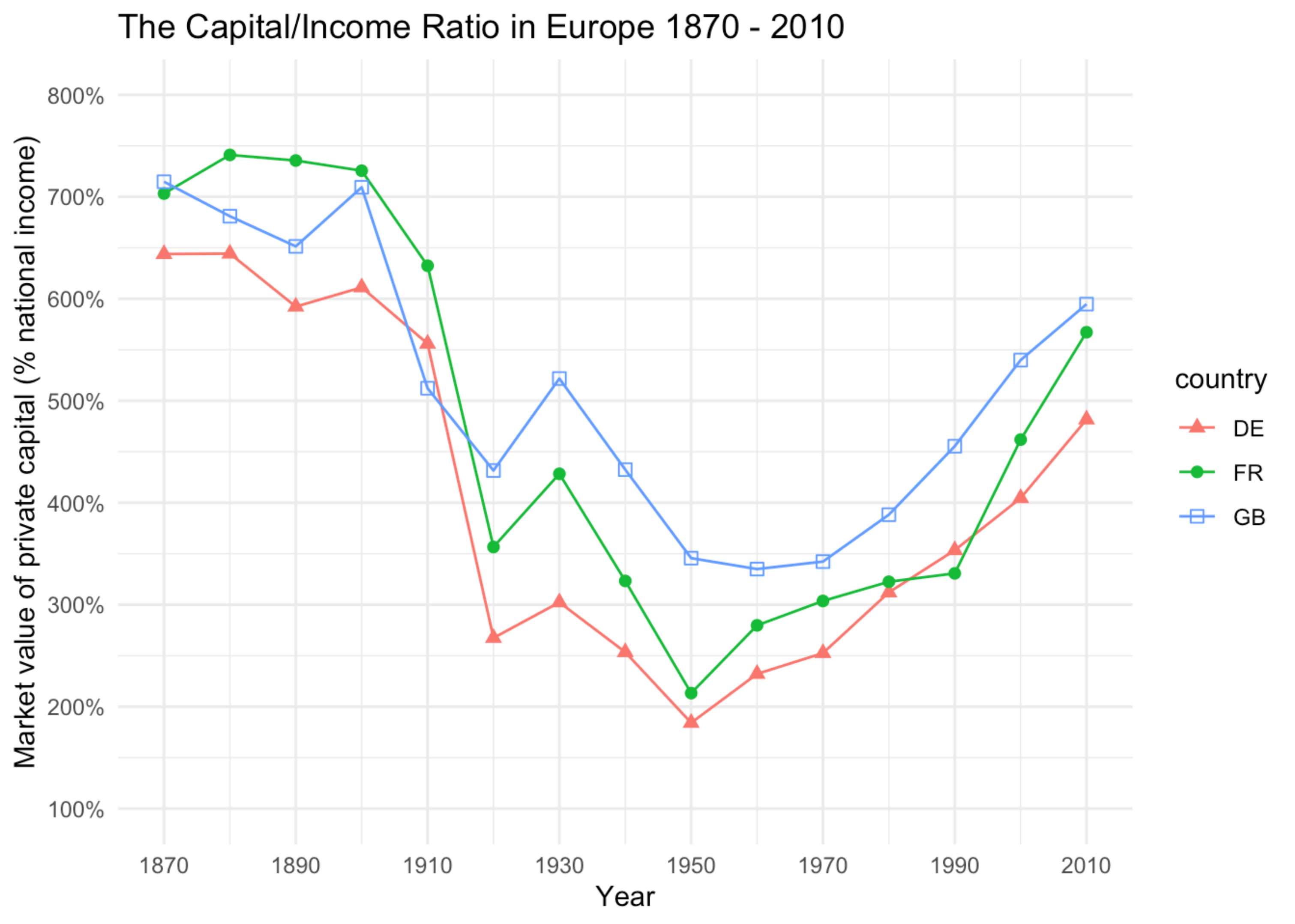
g

Rate of economic growth
(how fast total/average income grows)

If capital income grows at r, and total income only at g, then whenever r > g, capital's share of the total pie rises over time — automatically, by arithmetic.

Piketty calls this capitalism's "central contradiction": left alone, free accumulation tends toward increasing concentration — not because of malice or extra effort, but because of simple compounding arithmetic. Imagine you own a slice of the economy's capital: if your slice earns more each year (r) than the whole pie is growing (g), your slice becomes a larger fraction of the pie over time.

Capital-income ratio in Europe, 1970–present

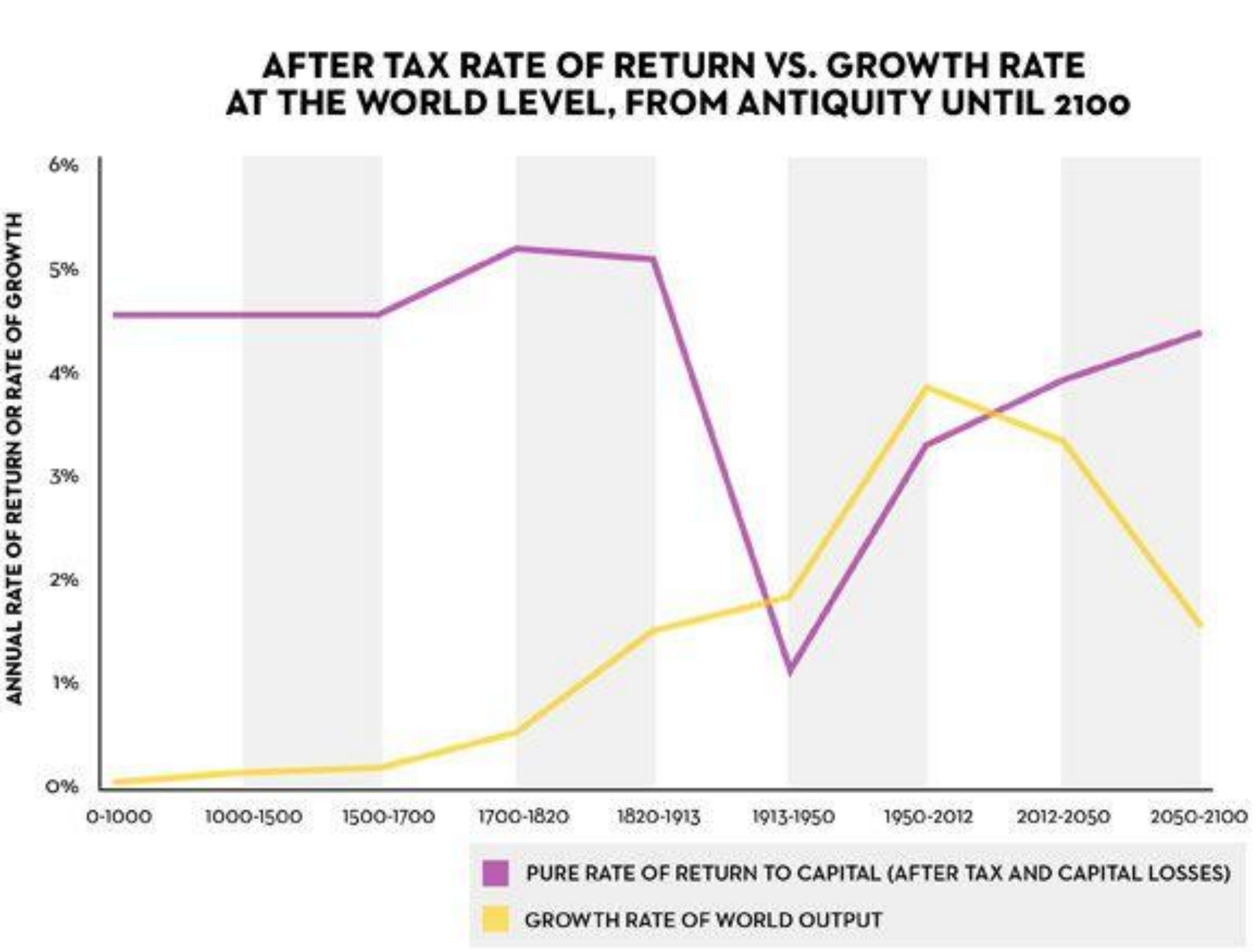


Capital: Total accumulated wealth or assets owned by a country, household, or company, E.g. real estate, financial assets, factories, machinery, foreign assets, etc.

Private capital-to-income ratios in rich countries fell sharply after WWI, WWII, and the Depression (capital destruction, inflation, nationalization).

They have been climbing back toward — or past — pre-WWI (Belle Époque) levels since the 1970s–80s.

The historical gap between r and g

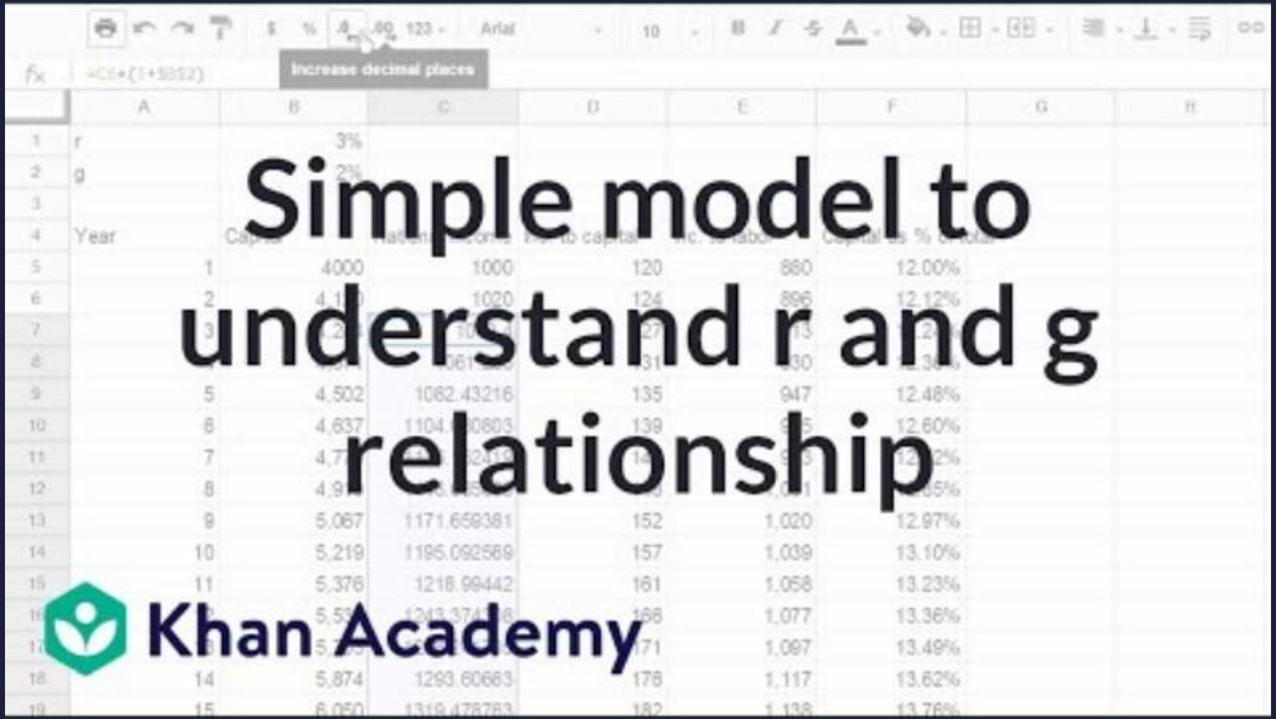


Historically, r has almost always exceeded g.

The major exception: roughly 1913–1980s, when wars, depressions, high taxation, and rapid postwar growth pushed g above or close to r.

Piketty's projection: as growth slows in the 21st century, r > g re-emerges as the dominant pattern — consistent with the rising inequality already shown.

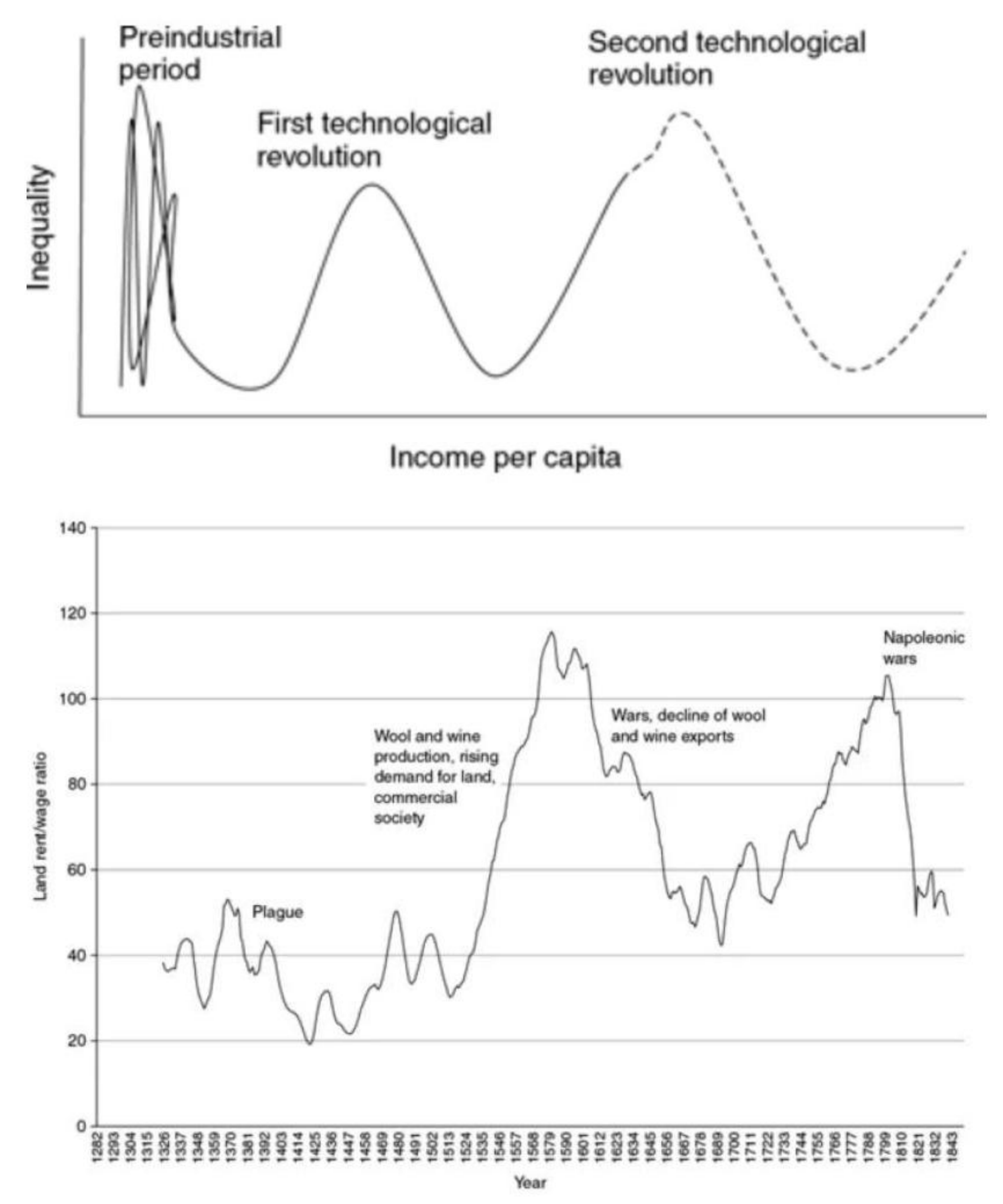
Demonstrating r > g with a simple example



https://www.khanacademy.org/economics-finance-domain/macroeconomics/macroeconomics-income-inequality/piketty-capital/v/piketty-spreadsheet-1

This short clip demonstrates, with a simple numerical example, how a gap between r and g compounds over time into rising wealth concentration.

Kuznets waves (Branko Milanovic)



Kuznets (1955): inequality rises then falls as economies industrialize — a single inverted-U.

Milanovic's revision: history shows repeated waves, not one curve. Falls are driven by "malign" forces (wars, revolutions, pandemics, state collapse) or "benign" forces (mass education, welfare states, progressive taxation, strong unions).

Ties back: the mid-20th-century dip in top shares (US, Europe, India) lines up with WWI/WWII, the Depression, and high wartime/postwar taxation — a wave trough, not an inevitable steady state.

PART 3

What does rising inequality imply?

Costs, counterarguments, and the debate over dynamism

Three costs of rising inequality

Social unrest

Perceived unfairness erodes social cohesion, fuels populism and polarization.

Slowing growth

Wealth concentrated among high-savers stagnates demand — money pools rather than circulates.

Harm to democracy

Concentrated wealth buys concentrated political influence — a feedback loop that entrenches itself.

Entrepreneurship vs. monopoly power

THE COUNTERARGUMENT

Inequality as the price of dynamism

Some inequality reflects legitimate returns to entrepreneurship, risk-taking, and innovation. Unequal outcomes can incentivize productive effort — flattening all returns could blunt innovation.

THE COUNTER-COUNTERARGUMENT

Accumulation breeds monopoly

$r > g$ shows accumulation doesn't require continued effort — capital compounds on its own. Markets tend toward monopoly absent active antitrust intervention: large capital holders out-compete or acquire new entrants, suppressing the very dynamism the entrepreneurship argument celebrates.

TWO RELATED CONCEPTS

Accumulation by dispossession & neoliberalism as a political project

Accumulation by dispossession

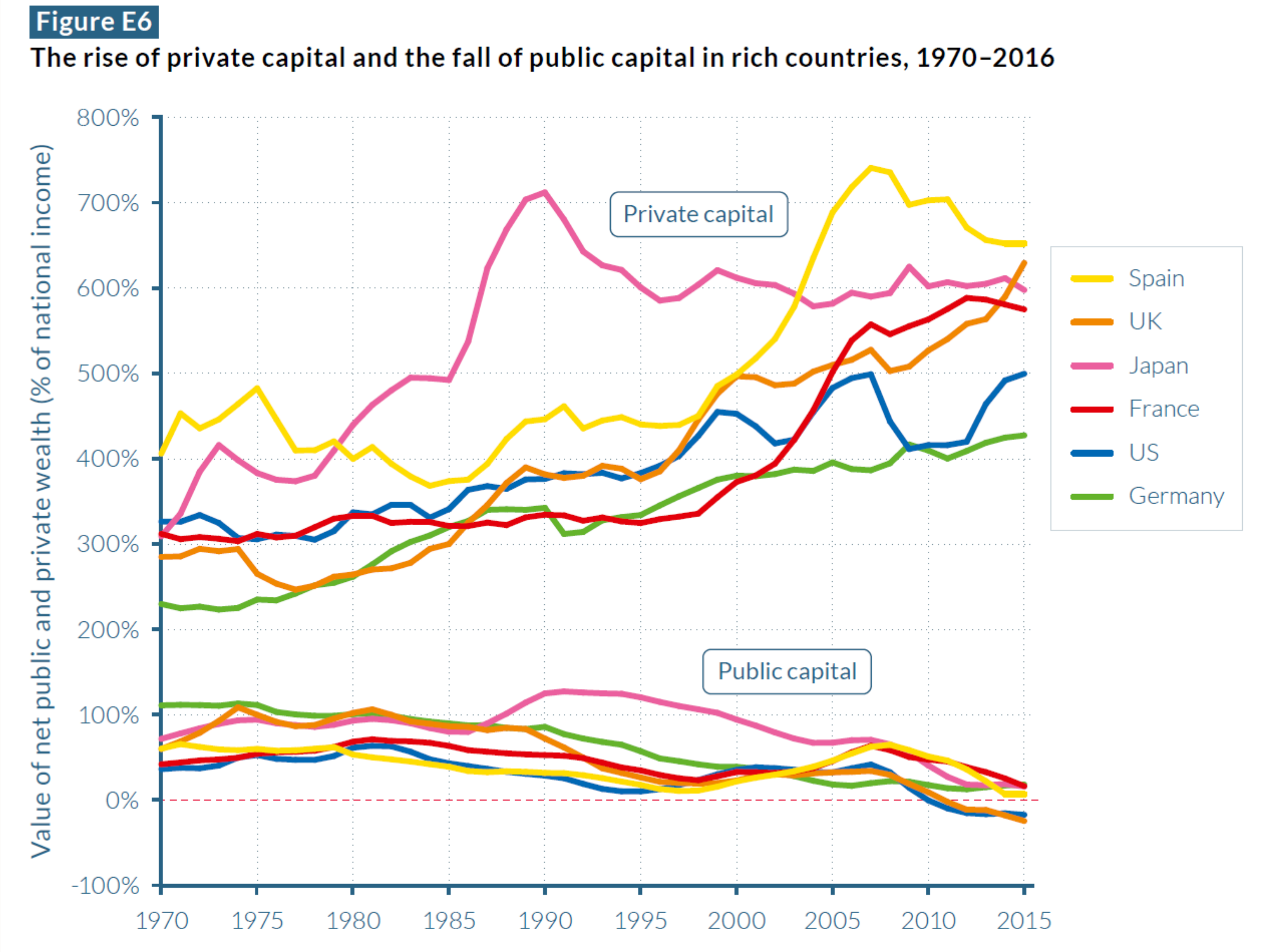
David Harvey's term: capital expands not just through new production but by transferring existing public/common assets into private hands — privatizing utilities, land, and natural resources. A redistribution upward, not new wealth creation.

Neoliberalism as a political project

Deregulation, privatization, and tax cuts are often framed technocratically — "efficiency," "market discipline" — but function as a political project shifting the balance between private and public wealth. Governments have gotten poorer as private wealth surged (next slide).

Common critique: framing a political/distributional choice as a purely technical one obscures who wins and who loses.

Governments got poorer as private wealth surged



1970 → today

Net private wealth: 200–350% → 400–700% of national income

Net public wealth: flat or negative in the US and UK

This is the concrete, measurable form that accumulation by dispossession and r > g have taken over the last half-century — wealth shifted from public to private hands, even as capital's own compounding advantage did the same from a different direction.

SUMMARY

Where this leaves us

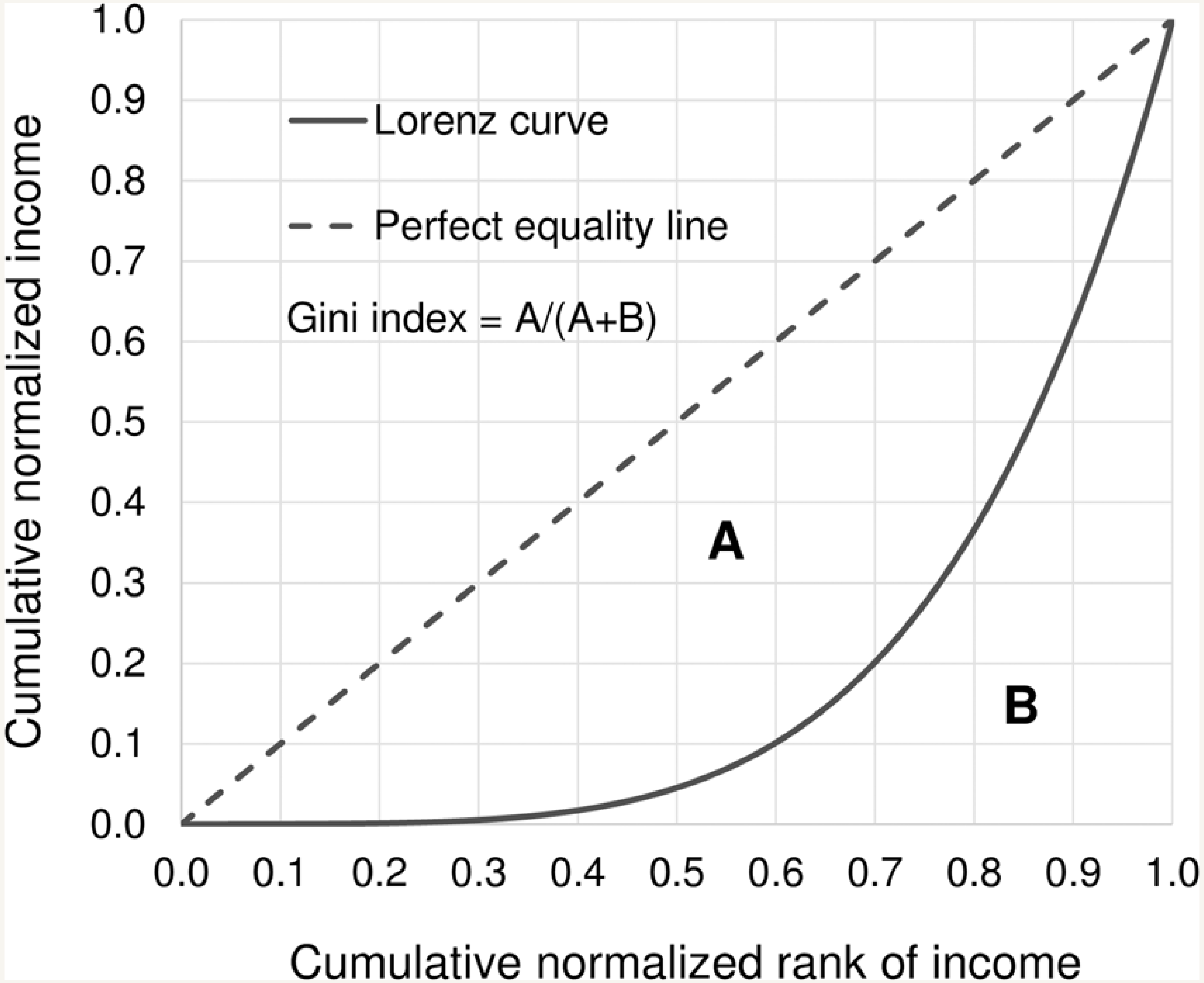
- Inequality has risen across almost all regions since 1980, at different speeds — institutions and policy explain the variation.
- India mirrors the global pattern: mid-century equalization, then a sharp post-1980s reversal, now at record highs.
- Piketty's r > g offers a structural mechanism for why capital-based inequality compounds over time.
- History shows this isn't inevitable — inequality moves in waves shaped by wars, social movements, and political choice.
- Rising inequality carries real economic, social, and democratic costs — but the trade-off with dynamism is a genuine debate worth engaging honestly.

A DEEPER LOOK

How do we measure inequality?

The Gini index is only the beginning of the story

The Lorenz curve



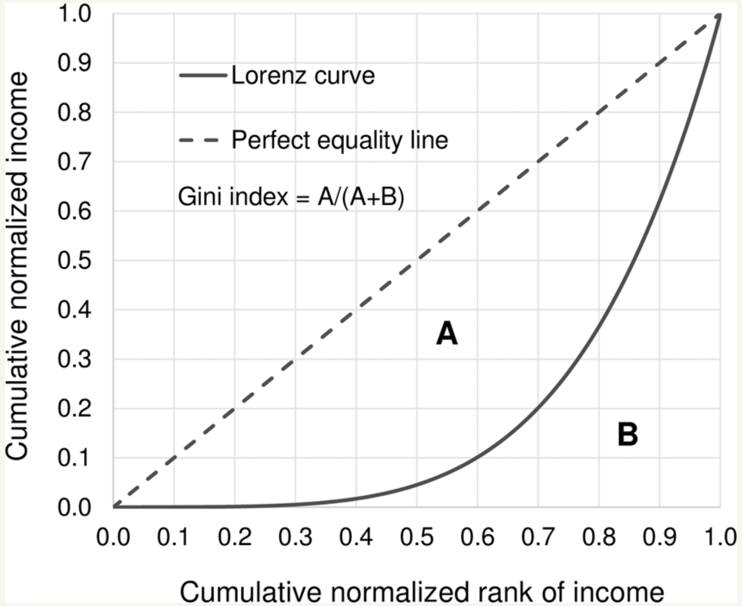
Plots the cumulative share of income (y-axis) held by the poorest x% of the population (x-axis).

The 45° line is perfect equality.

The further the curve bows below that line, the more unequal the distribution.

Gini, Atkinson, and Generalized Entropy are all, in different ways, built from this one curve.

The Gini index



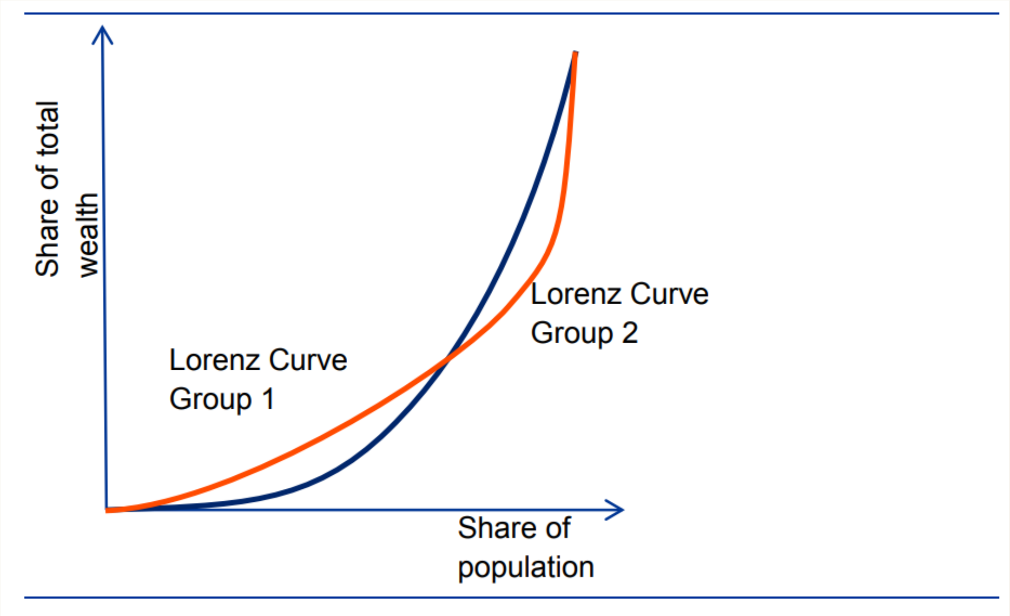
The ratio of the area between the 45° line and the Lorenz curve (A), to the total area under the 45° line (A + B).

Ranges from 0 (perfect equality) to 1 (perfect inequality).

Known weaknesses

- Less sensitive to inequality at the tails (very top and very bottom) — most sensitive to the middle of the distribution.
- If two countries' Lorenz curves cross, the Gini index can't reliably rank them.
- Two countries can share the same Gini index yet have very different gaps between rich and poor.

Same Gini index — very different income gaps



Greece

Gini = 0.360
Top10% / Bottom10% ratio = 13.8

Thailand

Gini = 0.360
Top10% / Bottom10% ratio = 8.9

Same Gini index. But the gap between the richest and poorest 10% is over 50% wider in Greece than in Thailand — a difference the Gini index alone cannot see.

Source: World Bank data, 2015, via Sitthiyot & Holasut (2020).

Four properties of a well-behaved measure

Anonymity

Unchanged if individuals are swapped — only wealth matters, not identity.

Population principle

Unchanged if the whole population is replicated.

Principle of transfers

The Pigou–Dalton test, from the previous slide.

Scale invariance

Unchanged if everyone's wealth is scaled by the same factor.

The Pigou–Dalton principle

A transfer of income from a richer person to a poorer person — as long as it doesn't make the poorer person richer than the (formerly) richer one — should never increase measured inequality. Ideally, it should strictly decrease it.

In plain terms: take a little money from a rich person and give it to a poorer person — while the rich person is still richer afterward. Any sensible inequality measure should treat this as a move toward equality.

Named after Arthur Cecil Pigou (1912) and developed by Hugh Dalton (1920). This is the axiom every inequality measure is judged against — some satisfy it strictly, some only weakly, and some not at all.

What is social welfare?

A social welfare function combines everyone's individual wellbeing into a single measure of how society as a whole is doing.

Why this matters for measuring inequality: several inequality measures — most notably the Atkinson index — are built by asking a welfare question directly: *"If we equally shared society's income, how much could we give up and still keep the same level of social welfare?“*

Social welfare functions can take different forms, E.g. sum of log of individual incomes.

The Atkinson index

Social welfare function

For a population of N individuals with incomes $y_1, y_2, \ldots, y_N$, the function changes depending on your choice of the inequality aversion parameter (ϵ):

- **When $\epsilon \neq 1$:**

$$W = \sum_{i=1}^{N} \frac{y_i^{1-\epsilon}}{1-\epsilon}$$

- **When $\epsilon = 1$ (The logarithmic case):**

$$W = \sum_{i=1}^{N} \ln(y_i)$$

The parameter ε controls how much weight is placed on the bottom of the distribution — higher ε means more concern for the poorest.

Built on the "equally distributed equivalent income" (**y_{EDE}**) idea: the income level per person that, if shared equally, gives the same social welfare as the actual distribution.

Atkinson index

$$W(y_1, y_2, \ldots, y_N) = W(y_{EDE}, y_{EDE}, \ldots, y_{EDE})$$

$$A = 1 - \frac{y_{EDE}}{\mu}$$

actual average income (μ)

For $\epsilon \neq 1$: $y_{EDE} = \left[\frac{1}{N} \sum_{i=1}^{N} y_i^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$

For $\epsilon = 1$: $y_{EDE} = \exp\left[\frac{1}{N} \sum_{i=1}^{N} \ln(y_i) \right]$

\+ Gives a complete ranking of distributions, and makes the underlying welfare judgment explicit — useful for policy.

− That ranking can shift a lot depending on the choice of ε, which is a value judgment, not a fact about the data.

Generalized Entropy indices

$$\mathrm{GE}(\alpha) = \frac{1}{\alpha(\alpha - 1)}\left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{y_i}{\bar{y}}\right)^{\alpha} - 1\right]$$

undefined if there are zero incomes. GE(0) is referred to as the mean logarithmic deviation, which is defined as follows:

$$\mathrm{GE}(0) = \frac{1}{N}\sum_{i=1}^{N}\ln\left(\frac{\bar{y}}{y_i}\right) \tag{4}$$

GE(1) is known as the Theil inequality index, named after the author who devised it in 1967. The Theil index is defined as follows:

$$\mathrm{GE}(1) = \frac{1}{N}\sum_{i=1}^{N}\frac{y_i}{\bar{y}}\ln\left(\frac{y_i}{\bar{y}}\right) \tag{5}$$

A family of measures indexed by α, which controls sensitivity to different parts of the distribution — low α weights the bottom, high α weights the top.

Avoids picking a social welfare function — instead built for a clean mathematical property.

Key strength

Additively decomposable: total inequality splits cleanly into a within-group component and a between-group component.

Trade-off: still depends on choosing α, and the theoretical range is unbounded (0 to ∞), which is harder to interpret than a 0–1 scale.

The same data, different verdicts

Each measure has a "sweet spot" in the income distribution where it is most sensitive to change.

Bottom-sensitive

Atkinson with high ε, GE with low α (e.g. mean log deviation)

Middle-sensitive

The Gini index

Top-sensitive

GE with high α (e.g. Theil), top-share indicators

A simple composite inequality index

$\left(\frac{B_{10}}{T_{10}}\right)_{\mathrm{i}}$ be the ratio of the income share held by the bottom 10% (B_{10}) to the income share held by the top 10% (T_{10}) in country i. In addition, let $H_{\mathrm{i}} = 1 - \left(\frac{B_{10}}{T_{10}}\right)_{\mathrm{i}}^{\alpha}$, $0 \leq H_{\mathrm{i}} \leq 1$. The exponent alpha ($\alpha$) is a weight such that $\text{avg. Gini} = 1 - \left(\text{avg.}\left(\frac{B_{10}}{T_{10}}\right)\right)^{\alpha}$, where avg. Gini is the average value of all countries' Gini index

α can be calculated as $\dfrac{\ln(1-\text{avg. Gini})}{\ln\left(\text{avg.}\left(\frac{B_{10}}{T_{10}}\right)\right)}$.

Ensures that the two components are roughly balanced in scale. Values between 0.197 and 0.281. Choose 0.25 for ease of calculations.

What α controls: it sets how strongly H reacts to that ratio.

- **Higher α** → H sits closer to 1 → more weight effectively falls on the tail-ratio component, less on the Gini index.
- **Lower α** → H sits closer to 0 → more weight effectively falls on the Gini index, less on the tail ratio.

$$I_{\mathrm{i}} = f(\text{Gini}_{\mathrm{i}}, H_{\mathrm{i}}) = \frac{\sqrt{\text{Gini}_{\mathrm{i}}^2 + H_{\mathrm{i}}^2}}{\sqrt{2}}, \quad 0 \leq I_{\mathrm{i}} \leq 1$$

Sitthiyot & Holasut (2020) combine two ingredients that each cover a blind spot of the other:

- The Gini index — sensitive to the middle of the distribution.
- The ratio of income held by the bottom 10% to the top 10% — sensitive to the tails.

Combined geometrically (via the Pythagorean theorem) into a single 0–1 index that responds to both.

Source: Sitthiyot & Holasut (2020), "A Simple Method for Measuring Inequality," Eq. (7).

Distinguishing countries the Gini index treats as identical

Country	Gini index	Composite index (I)	T10/B10 ratio
Greece	0.360	0.425	13.8
Thailand	0.360	0.391	8.9

Where the Gini index sees no difference, the composite index correctly ranks Greece as more unequal than Thailand — matching the T10/B10 evidence.

The same approach also distinguishes countries that share the same T10/B10 ratio but differ in their Gini index, and can detect rising inequality even when a country's Gini index looks flat over time.

Source: World Bank data, 2015, via Sitthiyot & Holasut (2020), Table 1.

SUMMARY

No single measure tells the whole story

- The Gini index is simple and widely reported, but under-weights the tails and can't rank distributions whose Lorenz curves cross.
- The Atkinson index makes a welfare judgment explicit, but its ranking depends on a subjective inequality-aversion parameter.
- Generalized Entropy indices decompose cleanly by subgroup, but remains sensitive to the choice of parameter and are unbounded.
- Combining a middle-sensitive and a tail-sensitive measure — as Sitthiyot & Holasut propose — can capture more of the picture in one number.
- The right choice depends on what you care about: the middle class, the poorest, the richest, or the ability to break inequality down by group.